



GOVERNMENT POLYTECHNIC, AMRAVATI
(AN AUTONOMOUS INSTITUTE OF GOVERNMENT OF MAHARASHTRA)
CURRICULUM DEVELOPMENT CELL

**PROGRAMME TITLE: DIPLOMA IN COMPUTER ENGINEERING/
INFORMATION TECHNOLOGY**

COURSE CODE: FC5490

**COURSE TITLE: EMERGING TRENDS IN COMPUTER ENGINEERING/
INFORMATION TECHNOLOGY(For Academic year 2020-2021)**

TEACHING SCHEME:

LEVEL OF COURSE	PRERE- QUISITE	WEEKLY CONTACT HRS.			TOTAL CREDITS	TOTAL WEEKS	TOTAL CONTACT HOURS		
		L	T	P			L	T	P
V		02	--	01	03	16	32	--	16

EXAMINATION SCHEME:

THEORY(Marks)					PRACTICAL(Marks)		TOTAL (Marks)
ESE PAPER HRS.	ESE		PA	TOTAL	ESE	PA	
02	MAX.	35\$	15*	50	25#	25^	100
	MIN.	14		20	10	10	-

@: Internal Assessment #: External Assessment Skill based \$: online examination

(*) Under the Theory PA, Out Of 15 Marks, 10 Marks is the Average of Two Tests and 05 Marks are for Micro project-

(^) Under practical PA Continuous Assessment of Practical Work is to be done by Course Teacher as per CDC norms.

For the courses having only practical examination, PA has two parts (i) Continuous Assessment of Practical work - 60% and (ii) -40%.

1. RATIONALE:

Advancements and Applications of Computer engineering and Information Technology are ever changing Emerging trends aims at creating awareness about major trends that will define technological disruption in the upcoming years in the field of Computer engineering and Information Technology. There are some emerging areas expected to generate revenue, increasing demand as IT professionals and open avenues of entrepreneurship.

2. COURSE OUTCOMES (COs)

At the end of this course, student will be able to: -

1. Describe Artificial Intelligence ,machine learning and deep learning
2. Interpret Internet of Things concepts.
3. Describe Ethical Hacking process and Detect Network Vulnerabilities..

3. DETAILED CONTENTS: THEORY

Unit	Unit Outcomes	Topics and Subtopics	CO No.	Marks	Hours
Unit-1 Artificial Intelligence	1a. Describe the concept of AI 1b. State the components of AI 1c. List Applications of AI 1d. Differentiate between machine learning & deep Learning	1.1 Introduction of AI Concept Scope of AI Components of AI Types of AI Application of AI 1.2 Concept of Machine Learning and Deep Learning	1	07	06
Unit 2 Internet of Things	2a. State the domains and application areas of Embedded Systems 2b. Describe IoT systems in which information and knowledge are inferred from data 2c. Describe designs of IoT 2d. State IoT Issues and challenges in deployment	2.1 Embedded Systems: Embedded system concepts. Purpose of Embedded system, Architecture of Embedded system, Types of Embedded processors–PIC, ARM, AVR, ASIC 2.2 Internet of Things : Definition and Characteristics of IoT Physical design of IoT Things of IoT IoT Protocols Logical design of IoT, IoT functional Blocks, IoT communication models, IoT communication APIs, IoT Enabling Technologies IoT levels and Deployment Templates, IoT Issues and Challenges, Applications IoT devices and its features: Arduino, Uno, Raspberry Pi, Node Microcontroller unit	2	14	14
Unit 3 Basics of Hacking	3a. define Hackers 3b. Describe the need of hack your systems 3c. Describe the dangers in systems 3d. Describe the Ethical Hacking process 3e. Describe Network Infrastructure Vulnerabilities	3.1 Ethical Hacking Defining hacker, Malicious users 3.2 Need to hack your own system. 3.3 Mention the dangers your systems face Nontechnical attacks Network-infrastructure attacks Operating-system attacks Application and other specialized attacks 3.4 Obeying the Ethical hacking Principles Working ethically	3	14	12

		Respecting privacy Not crashing your systems 3.5 The Ethical hacking process Formulating your plan Selecting tools Executing the plan Evaluating results Moving on 3.6 Network Hacking 3.6.1 Network Infrastructure: Network Vulnerabilities Scanning-Ports Ping sweep Scanning SNMP Grabbing Banners Analyzing Network Data and Network Analyzer MAC-daddy attack			
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4.LIST OF PRACTICALS:

Sr No.	PRACTICAL OUTCOMES (PrOs)	CO NO.
1.	Write a simple program using PROLOG.	1
2	Write a program to transfer serial data on your thingspeak account using Node MCU	2
3	Write a program to transfer serial data on your thingspeak account using Raspberry Pi	2
4	Write a program to transfer data to thingspeak cloud using Interface LM35 Temperature sensor to Node MCU8266	2
5	Use of network reconnaissance tools like WHOIS, dig, traceroute, nslookup to gather information about networks and domain registers	3
6	Download and install nmap. Use it with different options to scan open ports, perform OS fingerprinting, do a ping scan, tcp port scan, udp port scan, etc.	3

Note

- The entire above listed practical's need to be performed compulsorily, so that the students reach the 'Precision level of Dave's Psychomotor Domain'.
- The 'Process' and 'Product' related skills associated with each PrO are to be assessed according to a suggested sample given below.

S. No.	Performance Indicators	Weightage in %
1	Preparation of experimental setup	20
2	Setting and operation	20
3	Follow safety precautions	10
4	Observations and Recording	10
5	Interpretation of result and Conclusion	20

S. No.	Performance Indicators	Weightage in %
6	Answer to sample questions	10
7	Submission of report in time	10
Total		100%

The above PrOs also comprise of the following social skills/attitudes which are Affective Domain Outcomes (ADOs) that are best developed through the laboratory/field based experiences:

- Work collaboratively in team.
- Follow ethical Practices

The ADOs are not specific to any one PrO, but are embedded in many PrOs. Hence, the acquisition of the ADOs takes place gradually in the student when s/he undertakes a series of practical experiences over a period of time. Moreover, the level of achievement of the ADOs according to Krathwohl's 'Affective Domain Taxonomy' should gradually increase as planned below:

- 'Valuing Level' in 1st year
- 'Organization Level' in 2nd year and
- 'Characterization Level' in 3rd year

5. SUGGESTED STUDENT ACTIVITIES

Other than the classroom and laboratory learning, following are the suggested student-related **Co-curricular** activities which can be undertaken to accelerate the attainment of the various Outcomes in this course: Students should conduct following activities following activities in group and prepare reports of about 5 pages for each activity, also collect/record physical evidences for their (students) portfolio which will be useful for their placement interviews:

a. Prepare report on suggestive case study of Hacking as given below:

i.) The Aaron Cafferey case-United Kingdom ,2003

<http://digitalcommons.law.scu.edu/cgi/viewcontent.cgi?article=1370&context=chtlj>

ii) The Julie Amero case-Connecticut,2007

<http://dfir.com.br/wp-content/uploads/2014/02/julieamerosummary.pdf>

iii) The Michael Fiola case-Massachusetts,2008

<http://truthinjustice.org/fiola.htm>

b. Prepare report on any given case study of IoT..

6. SUGGESTED INSTRUCTIONAL STRATEGIES

These are sample strategies, which the teacher can use to accelerate the attainment of the various learning outcomes in this course:

- Massive open online courses (**MOOCs**) may be used to teach various topics/subtopics.
- The teacher needs to ensure to create opportunities and provisions for **co-curricular** Activities.
- About **10-15% of the topics/sub-topics** which is relatively simpler or descriptive in nature is to be given to the students for **self-directed learning** and assess the development of the LOs/COs through classroom presentations (see implementation Guideline for details).
- Procure various materials required for practical exercises
- Guide student(s) in undertaking micro-projects.
- Demonstrate students thoroughly before they start doing the practice.
- Observe continuously and monitor the performance of students in Lab.

7. SUGGESTED MICRO-PROJECTS.

Only one micro-project is planned to be undertaken by a student assigned to him/her in the beginning of the semester. She/he ought to submit it by the end of the semester to develop the industry oriented COs. Each micro-project should encompass two or more COs which are in fact, an integration of practicals, cognitive domain and affective domain LOs.

The microproject could be industry application based, internet-based, workshop-based, laboratory-based or field-based. Each student will have to maintain dated work diary consisting of individual contribution in the project work and give a seminar presentation of it before submission. The total duration of the

micro-project should not be less than **16 (sixteen) student engagement hours** during the course. In the all semesters, the micro-project could be group-based(4 to 5 students) to build up the skill and confidence in every student to become problem solver so that she/he contributes to the projects of the industry.

A Suggestive list of micro-projects is given here. Similar micro-projects could be added by the concerned faculty:

a.IoT Based Humidity and Temperature Monitoring

- i. The need of IoT Based Humidity and Temperature Monitoring .
- ii. The hardware requirements for designing this system.
- iii. The software requirements for designing this system.
- iv. Describe the circuit designed for this system along with its working
- v. Design an IoT application and mention how to store and retrieve a data on it.

b.IoT based Weather Monitoring System

- i. The need of IoT Based Weather Monitoring System
- ii. The hardware requirements for designing this system
- iii. The software requirements for designing this system.
- iv. Describe the circuit designed for this system along with its working
- v. Design an IoT application and mention how to store and retrieve a data on it .

c. Study Credit card fraud as an identity threat. Identify

- i. Use of digital media in carrying out fraud.
- ii. Vulnerability Exploited.
- iii. Effect of fraud.
- iv. Protection /Precaution to be taken against such frauds.

d. Study any Trojan attack. Identify the Trojan attack.

- i. State the way Trojan got installed on particular Machine.
- ii. State the effects of the Trojan
- iii. Elaborate/ Mention /State protection /Blocking mechanism for this Specific Trojan, Example specification of any anti-threats platform which filters the Trojan.

8. MAJOR EQUIPMENTS/INSTRUMENTS REQUIRED

Sr No.	Equipment Name with Broad Specification	Practical No.
1	PROLOG	1
2	Node MCU and Peripherals (e.g. LEDs, Sensors like Temp. sensor LM35, Ultrasonic sensor etc.)	2,3,4
3	Raspberry Pi	2,3,4
4	Arduino IDE (Open source software)	2,3,4
5	Python Software	2,3,4
6	Kali Linux	5,6
7	WHOIS, dig, traceroute, nslookup tools	5
8	n-map tool	5

9. SUGGESTED SPECIFICATION TABLE FOR QUESTION PAPER DESIGN

Unit No.	Unit Title	Marks per Unit	Distribution of Theory Marks		
			R Level	U Level	A Level
1	Artificial Intelligence	07	3	2	2
2	Internet of Things	14	4	6	4
3	Basics of Hacking	14	4	6	4
Total		35	11	14	10

R= Remember, **U**= Understanding, **A**=Application and above (*Bloom's Revised taxonomy*)

Note: This specification table provides general guidelines to assist student for their learning and to teachers to teach and assess students with respect to attainment of **UOs**. The actual distribution of marks at different taxonomy levels (of R, U and A) in the question paper may vary from above table

10. SUGGESTED LEARNING RESOURCES:

Sr.No.	Title Of Book	Author	Publication
1.	Artificial Intelligence	R.B.Mishra	PHI Learning Pvt. Ltd
2.	Introduction to Embedded Systems	Shibu K.V.	Tata Mcgraw Hill ISBN 978-0-07-014589-4
3.	Internet Of Things –A Hands –on Approach	Arshadeep Bahga,Vijay Madiseti	McGraw Hill Education, Private Limited,New Delhi,2010,ISBN:978-0-07-068178-1
4.	Hacking for Dummies	Kevin Beaver CISSP	Wiley Publishing Inc. ISBN:978-81-265-6554-2

11. SOFTWARE/LEARNING WEBSITES.

- <https://www.allitebooks.in/the-internet-of-things/>
- <https://www.versatek.com/wp-content/uploads/2016/06/IoT-eBook-version5.pdf>
- https://www.tutorialspoint.com/internet-of-things/intrnet-of-thins_tutorial.pdf
- <http://www.spmkck.co.in/Notes/Learning%20Internet%20of%20Things.pdf>
- <https://docs.microsoft.com/en-us/sysinternals/downloads/psloggedon>
- www.openwall.com/passwords/windows-pwdump
- https://www.tutorialspoint.com/ethical_hacking/ethical_hacking_process.htm
- <https://slideplayer.com/slide/7480056>

12. COURSE CURRICULUM DEVELOPMENT COMMITTEE:

SR. NO.	NAME	DESIGNATION	INDUSTRY/INSTITUTE
1.	Dr.P.P.Karde	H.O.D Information Technology	Govt. Polytechnic Amravati
2.	Dr. P R Satav	Lecturer in Computer Engineering	Govt. Polytechnic Amravati
3.	A.P.Jane	Lecturer in Information Technology	Govt. Polytechnic Amravati
4.	R.R.Bhoge	Lecturer in Information Technology	Govt. Polytechnic Amravati

Govt. Polytechnic, Programme Board of Studies of Computer Engineering has approved the above course curriculum on 30/12/2020 and is adopted for Computer Engineering Programme.

CHAIRMAN
PROGRAMME BOARD OF STUDIES,
COMPUTER ENGINEERING
GOVERNMENT POLYTECHNIC,AMRAVATI.

The General Board of Studies has approved the above course curriculum on 06/02/2021

The Governing Body has approved the above course curriculum on 13/08/2021

10.5.2 Elective Courses



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CURRICULUM DEVELOPMENT CELL

**PROGRAMME TITLE: DIPLOMA IN COMPUTER ENGINEERING /
INFORMATION TECHNOLOGY**

COURSE CODE: FC5471

COURSE TITLE: CLIENT SIDE SCRIPTING USING JAVASCRIPT

TEACHING SCHEME:

LEVEL OF COURSE	PRERE- QUISITE	WEEKLY CONTACT HRS.			TOTAL CREDITS	TOTAL WEEKS	TOTAL CONTACT HOURS		
		L	T	P			L	T	P
V	FC3408	03	-	02	05	16	48	-	32

EXAMINATION SCHEME:

THEORY(Marks)					PRACTICAL(Marks)		TOTAL (Marks)
ESE PAPER HOURS	ESE		PA	TOTAL	ESE	PA	
3Hrs	MAX.	70	30*	100	25#	25^	150
	MIN.	28	---	40	10	10	

@: Internal Assessment #: External Assessment: Practical Based

(*) Under the Theory PA, Out Of 30 Marks, 20 Marks is the Average of Two Tests and 10 Marks are for Micro project-

(^) Under practical PA Continuous Assessment of Practical Work is to be done by Course Teacher as per CDC norms.

For the courses having only practical examination, PA has two parts (i) Continuous Assessment of Practical work - 60% and (ii) microproject-40%.

1.RATIONALE:

JavaScript is limited featured client side programming language. JavaScript runs at the client end through the user's browser without sending messages back and forth to the server. It is widely used by the web developers to do things such as build dynamic web pages, respond to event, create interactive forms, validate data that the visitor enters into a form, control the browsers etc. This course helps student to create highly interactive web application using these features.

2. COURSE OUTCOMES (COs)

At the end of this course, student will be able to: -

1. Create interactive web pages using program flow control structure.
2. Implement Arrays and functions in JavaScript.
3. Create event based web forms using JavaScript.
4. Use JavaScript for handling cookies.
5. Create interactive web page using regular expression for validation.
6. Create menus and navigations in web pages.